

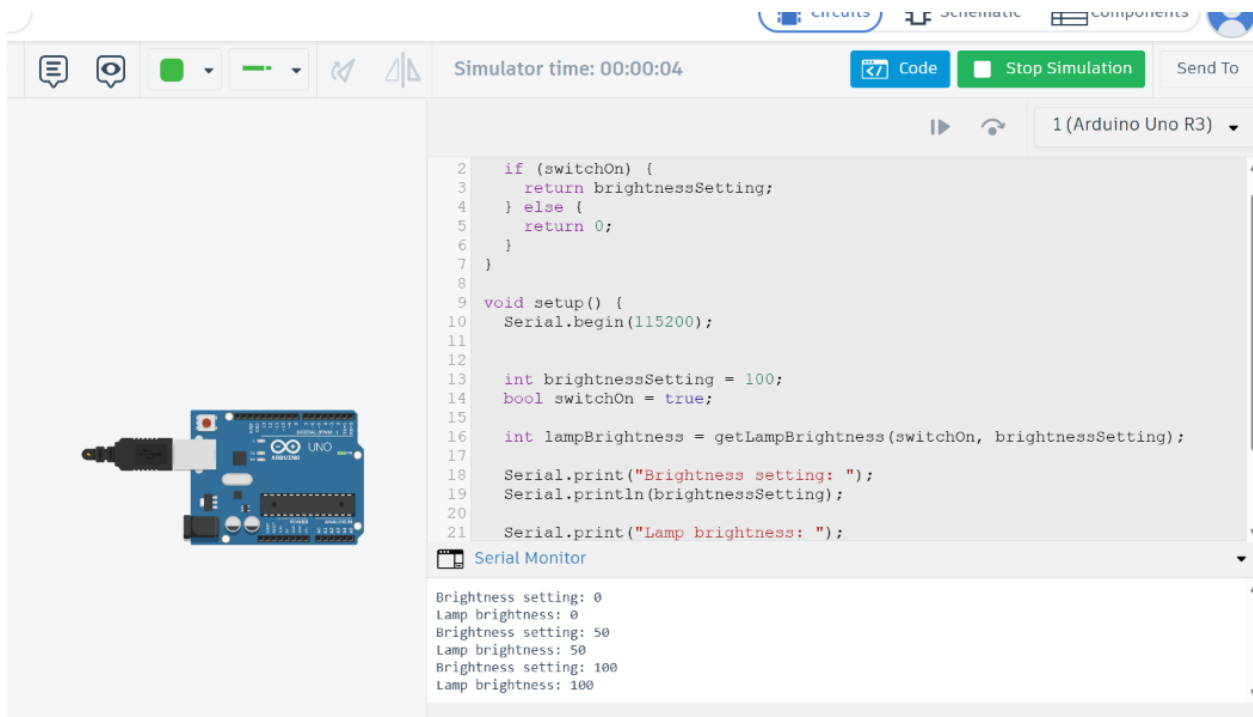
Laboratory Activity 2

Brightness Setting	Switch State	Expected Lamp Brightness	Observed Output
0	On	0	Brightness setting: 0 Lamp brightness: 0
50	On	50	Brightness setting: 50 Lamp brightness: 5
100	On	100	Brightness setting: 100 Lamp brightness: 100
0	Off	0	Brightness setting: 0 Lamp brightness: 0
50	Off	0	Brightness setting: 50 Lamp brightness: 0
100	Off	0	Brightness setting: 100 Lamp brightness: 0

Explanation of Parameter and Return Value

- **bool switchOn (Parameter):** Accepts a Boolean value (true or false) representing whether the lamp switch is flipped on or off.
- **int brightnessSetting (Parameter):** Accepts an integer (typically 0 to 100) representing the user's configured target brightness level.
- **Return Value (int):** Produces the effective brightness level evaluated by the control logic. If switchOn evaluates to true, it returns the value passed into brightnessSetting. If switchOn is false, it returns 0, simulating an unlit lamp without modifying the original setting variable.

Output:



Simulator time: 00:00:04

Code

Stop Simulation

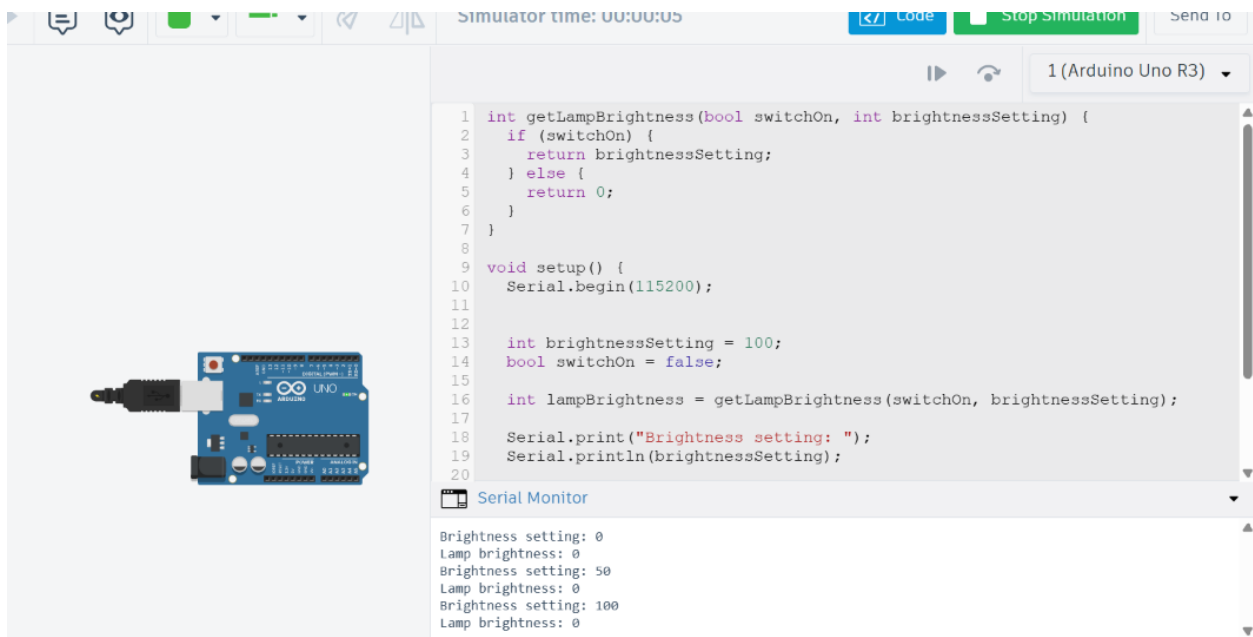
Send To

1 (Arduino Uno R3)

```
2   if (switchOn) {
3       return brightnessSetting;
4   } else {
5       return 0;
6   }
7 }
8
9 void setup() {
10    Serial.begin(115200);
11
12    int brightnessSetting = 100;
13    bool switchOn = true;
14
15    int lampBrightness = getLampBrightness(switchOn, brightnessSetting);
16
17    Serial.print("Brightness setting: ");
18    Serial.println(brightnessSetting);
19
20    Serial.print("Lamp brightness: ");
```

Serial Monitor

Brightness setting: 0
Lamp brightness: 0
Brightness setting: 50
Lamp brightness: 50
Brightness setting: 100
Lamp brightness: 100



Simulator time: 00:00:05

Code

Stop Simulation

Send To

1 (Arduino Uno R3)

```
1 int getLampBrightness(bool switchOn, int brightnessSetting) {
2   if (switchOn) {
3       return brightnessSetting;
4   } else {
5       return 0;
6   }
7 }
8
9 void setup() {
10    Serial.begin(115200);
11
12    int brightnessSetting = 100;
13    bool switchOn = false;
14
15    int lampBrightness = getLampBrightness(switchOn, brightnessSetting);
16
17    Serial.print("Brightness setting: ");
18    Serial.println(brightnessSetting);
19
20    Serial.print("Lamp brightness: ");
```

Serial Monitor

Brightness setting: 0
Lamp brightness: 0
Brightness setting: 50
Lamp brightness: 0
Brightness setting: 100
Lamp brightness: 0

Coding Check

1. In `int brightness Setting = 75;` identify the data type, variable name, and initial value.

How would you change the value to 50?

Data type: `int`

Variable name: `brightness Setting`

Initial value: 75

Change value to 50: Reassign or initialize it as `int brightness Setting = 50;`

2. Why is `bool` suitable for `switch On`, while `int` is suitable for `brightness Setting`?

`bool switch On`: Suitable because a switch has only two binary states—true (on) or false (off).

`int brightness Setting`: Suitable because brightness is a range of whole numbers from 0 to 100.

3. Predict the output when `brightness Setting` is 80 and `switch On` is false.

Brightness setting: 80

Lamp brightness: 0

4. Are `switch On` and `switch on` the same variable? Explain.

No, they are not the same. C/C++ (and Arduino) is case-sensitive, so the uppercase O makes `switch on` a distinct identifier from `switch on`.

5. What values are passed to `getLampBrightness()`, and what does it return?

Values passed (parameters): Typically, the current switch state (`switchOn`) and the configured level (`brightnessSetting`).

Return value: Returns 0 if `switchOn` is false, or the numeric `brightnessSetting` value (e.g., 80) if true.

6. If `lamp Brightness` is declared inside `setup()`, can `loop()` access it directly? Explain.

No, `loop()` cannot access it directly. Variables declared inside `setup()` are local to that function's scope. To access it in both functions, declare it outside as a global variable or pass it as a parameter.